

COMPREHENSIVE REVIEW

Artificial intelligence and food flavor: How AI models are shaping the future and revolutionary technologies for flavor food development

Zhiyong Cui¹ | Chengliang Qi¹ | Tianxing Zhou^{1,2} | Yanyang Yu¹ |
Yueming Wang¹ | Zhiwei Zhang¹ | Yin Zhang³ | Wenli Wang¹ | Yuan Liu^{1,4} 

¹Department of Food Science & Technology, School of Agriculture & Biology, Shanghai Jiao Tong University, Shanghai, China

²Department of Bioinformatics, Faculty of Science, The University of Melbourne, Melbourne, Victoria, Australia

³Key Laboratory of Meat Processing of Sichuan, Chengdu University, Chengdu, China

⁴School of Food Science and Engineering, Ningxia University, Yinchuan, China

Correspondence

Yuan Liu, Department of Food Science & Technology, School of Agriculture & Biology, Shanghai Jiao Tong University, Shanghai 200240, China. Email: y_liu@sjtu.edu.cn

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Abstract

The food flavor science, traditionally reliant on experimental methods, is now entering a promising era with the help of artificial intelligence (AI). By integrating existing technologies with AI, researchers can explore and develop new flavor substances in a digital environment, saving time and resources. More and more research will use AI and big data to enhance product flavor, improve product quality, meet consumer needs, and drive the industry toward a smarter and more sustainable future. In this review, we elaborate on the mechanisms of flavor recognition and their potential impact on nutritional regulation. With the increase of data accumulation and the development of internet information technology, food flavor databases and food ingredient databases have made great progress. These databases provide detailed information on the nutritional content, flavor molecules, and chemical properties of various food compounds, providing valuable data support for the rapid evaluation of flavor components and the construction of screening technology. With the popularization of AI in various fields, the field of food flavor has also ushered in new development opportunities. This review explores the mechanisms of flavor recognition and the role of AI in enhancing food flavor analysis through high-throughput omics data and screening technologies. AI algorithms offer a pathway to scientifically improve product formulations, thereby enhancing flavor and customized meals. Furthermore, it discusses the safety challenges of integrating AI into the food flavor industry.

KEYWORDS

artificial intelligence, flavor perception, food design, molecular generation

1 | INTRODUCTION

Flavor is a key attribute of food, profoundly affecting all aspects of food processing, including production, trade, and intake. Good flavor is closely related to a happy mood, whereas unpleasant flavors, such as bitterness, often dis-

courage consumption (Maehashi & Huang, 2009). Due to improper eating habits, an unbalanced (overdone) diet is causing many people suffering from chronic diseases such as obesity, diabetes, heart disease, or stroke, which often lead to premature death (Lopez Barrera & Hertel, 2023). Furthermore, food waste is also a significant issue

resulting from bad taste, inadequate awareness of avoiding food waste, and so on (Gatto & Chepeliev, 2024). Food flavor directly affects consumers' dietary choices and consumption patterns. Enhancing food flavor can promote consumers to take a diverse and nutritionally balanced diet and reduce food waste.

With the improvement of large-scale computing models and hardware, large language models (LLMs), such as Llama (Touvron et al., 2023), PaLM (Singhal et al., 2023), and chatbot generative pre-trained transformer (ChatGPT) (Achiam et al., 2023), have demonstrated impressive results in natural language processing tasks, even achieving state-of-the-art performance on specific tasks. Interestingly, these models can still do some tasks effectively even when some data are not available (Martini et al., 2022). For example, LLMs have the capability to simplify scientific knowledge and provide personalized, site-specific, and data-driven agricultural advice, thereby improving agricultural extension capabilities (Tzachor et al., 2023). Since its release, LLMs have shown remarkable performance in “few-shot learning” or “zero-shot learning” tasks, which impressed scientists and generated numerous publications in their fields (Table S1) (Tian et al., 2024). However, the general applicability of these LLMs to unseen tasks in more complex domains, such as food flavor, has not been fully summarized and documented. The traditional identification and improvement of flavor substances is complicated, time-consuming, labor-intensive, and economically inefficient. It is necessary to apply accurate and efficient computer-assisted methods to identify flavor substances. Emerging computational strategies, such as multi-omics and artificial intelligence (AI), provide new opportunities to mine undiscovered natural flavor molecules from foods and design dedicated and safer artificial flavorings, which are effective complements to experimental methods (An et al., 2021). AI can skillfully collect and query large datasets, reveal complex patterns in high-dimensional variable spaces, and autonomously learn and generalize rules from new data, making it a powerful tool for accurately predicting and adjusting food flavors (Queiroz et al., 2024). By applying LLMs such as GPT-3.5 and GPT-4 to the study of molecular flavor properties, new research avenues are provided for sensory science, consumer science, and the food industry, which is of great significance for understanding consumer taste preferences and accelerating the development of new foods (Song et al., 2024).

The concept of using food as a form of medicine has been prevalent since ancient times, especially in the use of botanical drugs. It is widely accepted that the taste of these botanical drugs shaped their usage. In ancient Greece and Rome, the intended effects of plant-based medicines were often explained by their taste (Leonti et al.,

2023), which was also the case with traditional Chinese medicine and traditional Indian medicine. Although these treatment methods have historical basis and have been widely used locally, there is still a lack of reliable pathological basis, especially the evidence that aligns with modern medicine. However, research into the relationship between taste and biological activity is gradually progressing. For example, a linear correlation has been found between the effective concentration of a compound that inhibits IL-6 release caused by Pg-LPS and human's bitter taste threshold concentration, as published in the literature (Tiroch et al., 2023). Walter Willet et al. found a U-shaped relationship between carbohydrates intake and mortality, stating that the mortality rates increased when carbohydrates were replaced with animal-derived fat or protein (Seidemann et al., 2018). The FDA has approved 171 AI and machine learning (ML) medical devices dedicated to the development of digital health, containing mobile health, health information technology, wearable devices, telehealth and telemedicine, and personalized medicine. With the advancements in statistical methods and the accumulation of data, AI can use data from medicine, healthcare, and nutrition to predict personal disease risks and design personalized nutritional decisions for individual health improvement.

2 | PERCEPTION MECHANISM AND RAPID IDENTIFICATION OF FLAVOR SUBSTANCES

2.1 | Molecular mechanisms of flavor recognition and multisensory flavor perception

Flavor plays a crucial role in determining whether food can be accepted by consumers, with flavor receptors being a decisive factor in the perception of flavor. Flavor can be categorized into two main components: odor and taste (Policarpo et al., 2024). Odor perception mainly occurs through the olfactory receptor (OR) of GPCR family A, whereas taste perception involves various receptors for different tastes such as sourness (PKD1L2, PKD1L3 of OTOP1 proton channel), sweetness (T1R2–T1R3 of GPCR C family) (Mafi et al., 2021), bitterness (TAS2Rs of GPCR T family), saltiness (TCM4 of ENaC channel), umami (GRM4 (mGluR4) (Chaudhari et al., 1996), TAS1R1–TAS1R3, GRM1 (mGluR1), GPRC6A, CASR, GPR92 of GPCR family C) (Nelson et al., 2002). All these receptors except for the sour and salty taste receptors are GPCR proteins (Figure 1a). In addition, spiciness (actually a pain sensation, mediated by TRPV1 (H. Zhang, Lin, et al., 2023)), kokumi, fat (Besnard et al., 2016),

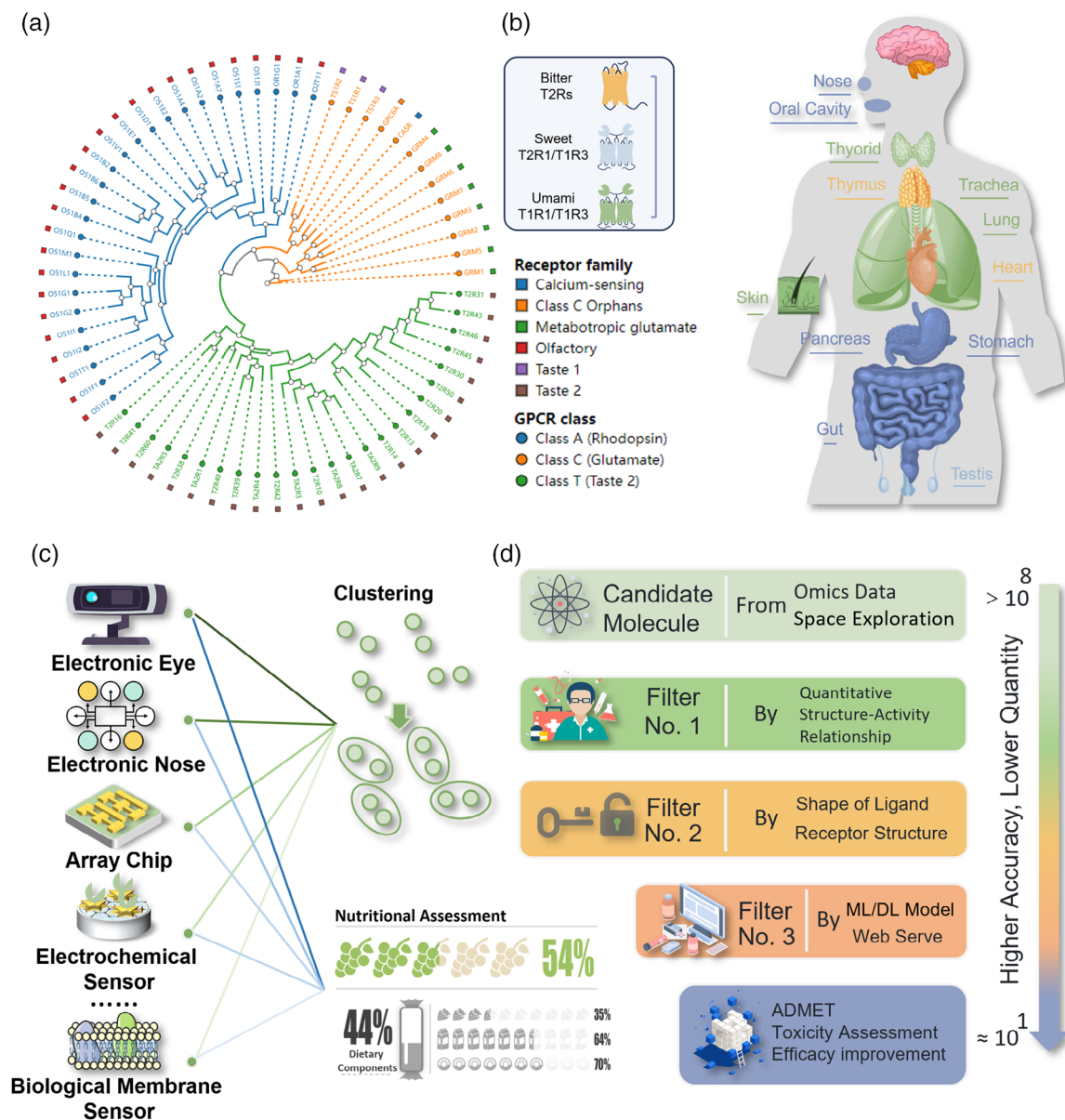


FIGURE 1 (a) Structural similarity tree diagram of flavor sensing proteins among GPCR proteins; (b) schematic diagram of the distribution of taste receptors. Taste receptors are widely distributed in multiple organs and tissues; (c) applications of common sensors in food ingredient clustering and nutritional assessment; (d) a multi-step-based strategy for screening flavors from potential chemical space. Source: (a) Drawn by GPCRdb36.

astringency, ammonium chloride (OTOP) (Liang et al., 2023), and metallic sensation (Pirkwieser et al., 2021) are often cited as non-standard flavors. It is the combination of these tastes and olfactory stimulation that gives rise to descriptors such as “fruity,” “meaty,” “floral,” and “herbaceous.” Beyond the chemical receptor-based senses of smell and taste, other senses, like hearing and vision, are also considered to be involved in the formation of integrated senses. The mechanism by which the brain integrates sensory information from multiple senses is called crossmodal correspondences (An et al., 2023; Zellner, 2013). For example, sound is considered a forgotten

flavor in some scenarios (e.g., famous British chef Heston Blumenthal) (Zampini & Spence, 2004), with the different characteristics of melody (balanced vs. unbalanced, smooth vs. jagged, symmetrical vs. asymmetrical, simpler vs. more complex) found to be linked with basic tastes (Velasco et al., 2023). Research by Erick G. Chuquichambi found that people generally associate curved and symmetrical shapes with sweetness, whereas angular and asymmetrical shapes are associated with other basic tastes like sour and bitter. Touch may also play a part in flavor perception, as shown in the “rubber hand illusion” experiment (Spence, 2015). In addition to the interaction among

the five senses, taste also has a significant impact on emotions. Some studies have reported that sour taste triggers the formation of disgust feelings (Zhang et al., 2019).

Increasing evidence also suggests that there are shared signaling pathways between flavor pathways, which provides a molecular explanation for the interaction of flavor components. Calcium homeostasis modulator 1/3 channel, a voltage-gated ion channel, has been shown to mediate GPCR neurotransmission of taste, including sweet, bitter (Taruno et al., 2013), umami (Ma et al., 2018), calcium taste, sodium taste (Nomura et al., 2020), and fat taste (Sclafani & Ackroff, 2018). Quinine, which has a bitter taste, interferes with the signal transduction of sweet taste by inhibiting TRPM5 (Talavera et al., 2008). Bitter substances, such as caffeine, induce proton secretion by interacting with bitter taste receptors (TAS2R), resulting in increased cytoplasmic Ca^{2+} and cAMP concentrations. Studies have found that both bitter compounds, caffeine and L-arginine, can induce TRPM4/M5-mediated Na^+ influx. After stimulating HGT-1 cells with the acetylcholine analog carbachol, TRPM4/M5 mediated Na^+ influx, indicating that the digestion-related cholinergic pathway interacts with the taste signaling pathway in parietal cells (Richter et al., 2024). Considering the above phenomena, it is not surprising that tastes interfere with each other. In addition to interference between signaling pathways, taste molecules may also have additional interactions with other taste receptors. Human sensory studies investigated taste interactions between sodium chloride (NaCl) and bitter tastants and observed an inhibitory effect of sodium on bitter taste. TAS2R16 signaling was reduced in the presence of NaCl. This suggests that for some receptor-agonist pairs, NaCl reduces activation at the receptor level, whereas central effects may dominate NaCl-induced inhibition of bitter taste for other substances (Kumar & Behrens, 2024). The crystal structure (PDB ID: 5 × 2 M) demonstrated that Cl^- binds near the side chain hydroxyl group of Thr105 and the main chain amide groups of Gln148 and Ser149, which might be involved in the activation of T1r2a/T1r3 LBD (Atsumi et al., 2023). In addition, Cl^- has also been observed to regulate the functions of mT1R2a/mT1R3 and zT1R2a/zT1R3, and their increased sensitivity to L-Pro-specific Cl^- dependency may lead to changes in zebrafish food preference due to environmental fluctuations (Goda et al., 2023). It is possible to study the recognition process between taste molecules and taste proteins through molecular simulation and chimera construction. However, the precise structure of taste receptors has not been fully resolved, so there is no direct structural evidence to explain the active site of taste receptors, resulting in the unclear binding mechanism between taste receptors and taste molecules. In summary, the interaction of flavor components is mediated by shared signaling pathways

and complex interactions among various taste receptors and molecules. The increasing evidence of these shared pathways provides a molecular explanation for the interaction of flavor components and highlights the intricate nature of taste perception and signal transduction. Further research into the precise structures and mechanisms of taste receptors will be crucial in fully understanding these interactions and their implications for taste perception and flavor science.

Flavor receptors, which are not only found on the tongue's taste buds but also distributed throughout various tissues and organs in the human body (Figure 1b), play a significant role in regulating physiological activities and maintaining nutritional balance (Dubovski et al., 2022). The above receptors exist in the gut-brain, gut-lung, gut-heart, gut-kidney, and gut-liver axes and are involved in physiological and biochemical activities such as systemic metabolic regulation, energy balance, and food intake (Li et al., 2024). L-glutamate promotes digestion by transmitting taste signals to the brain through receptors in the mouth or stomach, which, in turn, increases the secretion of saliva in the parotid glands and gastric juice (Kouzuki et al., 2019). In the respiratory tract, bitter taste receptors are expressed and serve an important role in protecting the airways from pathogens and diseases like bronchoconstriction. Studies have shown that bitter substance-induced airway smooth muscle relaxation is mediated by TAS2Rs-activated $\text{G}\alpha_t$ release, and the released $\text{G}\alpha_t$ blocks the contractile response by effectively inhibiting the $\text{G}\alpha_q$ cycle of AChR signaling (Zhou et al., 2022). Muscone, a compound that binds to the OR MOR215-1, activates the cAMP signaling pathway to initiate transgene expression, which has potential therapeutic implications for diseases like non-alcoholic fatty liver disease and allergic asthma (Wu et al., 2024). Furthermore, individual variations in GPCR sequences can affect taste perception and have been reported to be associated with food selection, preferences, and even disorders and diseases. For example, polymorphisms in TAS2Rs were found to be associated with chronic otitis media and affect fertility (Kaufman et al., 2021). A comprehensive understanding of the differential effects of ligands on naturally occurring taste GPCR variants may pave the way for a better grasp of individual differences, personalized nutrition, and the development of therapeutic strategies.

2.2 | Intelligent sensory and evaluation strategies

Due to the constraints of time, space cost, and repeatability in traditional sensory evaluation, researchers have been developing reliable, fair, accurate, repeatable,

and cost-effective sensory evaluators such as electronic noses/tongues/eyes. These advanced evaluators use various technologies, including lipid/polymer membranes, conjugated polymers, hydrogels, and human taste cells (Figure 1c), to mimic human sensory responses, enabling rapid analysis of aromas, flavors, and appearances. They enhance evaluation efficiency and consistency, propelling quality control and innovation in various industries like food, fragrance, and cosmetics (Jung et al., 2023). Miao et al. constructed hydrogel electronic tongue based on two different responses (electrical response and volumetric response) to identify salty/sour or sweet tastes. They used an ML algorithm to fit the signals from tastant-induced electrical and volume responses to achieve taste recognition (Miao et al., 2024). A multi-layer feedforward neural network (back propagation neural network) model based on the output signals of the electronic nose was proposed to well predict the flavor acceptability of yogurt (Tian et al., 2020). Q-Grader combined surface-engineered nano-hybrid transistor sensors with AI statistical analysis technology to simulate human sense of smell and taste. It used metal nanoparticles (Au, Ag, and Pd NPs@ZnO) as core sensing elements and constructs a sensor array with two ends crossed. It could effectively detect and distinguish the unique aroma of different types of coffee beans (such as decaffeinated beans from Papua New Guinea, Brazil, Ethiopia, and Colombia) (Jang et al., 2024). Similarly, Intel Corporation and Cornell University presented a neural algorithm for rapid online learning and identifying odorant samples under noise. The algorithm, based on the architecture of mammalian olfactory bulb, is implemented on the Intel Loihi neuromorphic system, which uses 72 chemical sensors to detect various odorants, functioning as an “electronic nose” to recognize odors as a reactive database (Imam & Cleland, 2020). Moreover, a reusable graphene field effect transistor conjugated with the TRPV1 nanodisk has been reported for assessing spicity-related pain, which can accurately identify 1 aM capsaicin, 10 aM dihydrocapsaicin, 1 fM piperine, 10 nM garlic, and 1 pM AITC (Seo et al., 2023). Another device, an all-polymer electrochemical transistor ionic tactile taste receptor (channel length 2 μm) has the capability to combine tactile and taste perception (G. Liu, Wen, et al., 2023). An electronic nose sensor array acquisition technology based on multiple overlapping sniffs can obtain time series signals with rich dynamic features, thereby improving the perception and identification capabilities of different gases. The bionic lung developed by Liu et al. can continuously inhale test gas at a frequency of 1 Hz and inject it into the surface of the catalytic chemiluminescence sensor, thereby realizing gas sampling with multiple overlapping sniffing strategies, and can produce cross-responses to volatile organic compounds in complex

gas components, thereby achieving differentiated perception of different gases (L. Liu, Na, et al., 2024). Although these are great advancements, current research on the interaction between taste and substances (synergistic and inhibitory effects) is still insufficient, and the problem of electronic sensors significantly deviating from natural taste also needs to be addressed. Although electronic tongues made of synthetic materials, such as polymers, semiconductors, or lipid membranes, have been used for taste detection, their low sensitivity and specificity limit their application and promotion. It is necessary to strengthen the development of more taste intensity assessment models and establish a unified taste intensity assessment standard.

2.3 | Flavor substance database

Flavor molecules in food initiate chemical reactions that produce taste and aroma, playing an important role in metabolic regulation. As the number of flavor molecule reports increases, alongside the updating in database algorithms and internet technology, more and more flavor database studies have been published. Table 1 lists the flavor databases, including their focus, uniform resource locator, data availability, and number of molecules. A total of 25,595 flavor molecules have been recorded on the FlavorDB 2 website, of which 2254 are derived from natural products and 13,869 are synthetic substances (Goel et al., 2024). The database provides information, including basic identity, functional groups, physicochemical properties, 2D/3D properties, and ADMET (absorption, distribution, metabolism, excretion, and toxicity) properties of each flavor molecule. Additionally, FlavorDB contains information on 33 taste receptors (sweet, bitter, sour, and umami) and 1068 odor receptors, with UniProt links providing more detailed information about each receptor (Garg et al., 2017). *ChemTastesDB* organizes 2944 verified and curated compounds into 5 basic taste categories (sweet, bitter, umami sour, and salty) and 4 non-basic taste categories (tasteless, non-sweet, multitaste, and miscellaneous) and records the name, PubChem CID, CAS registry number, canonical SMILES, class taste, and references of the flavor substances (Rojas et al., 2022). The only drawback is that the data of *ChemTastesDB* is only hosted on Zenodo without a portal website. *PhytoMolecularTasteDB* collects detailed information on plant-derived flavor molecules used in traditional Indian medicine and taste combinations produced by major flavor molecules in medicinal plants, including 431 Ayurvedic medicinal plants, 94 EPAs, 223 phytochemical classes, and 438 plant-derived taste substances (Dragos & Gilca, 2018). In recent years, taste peptides have attracted widespread attention

TABLE 1 Flavor database.

Name	Description (size)	Web site
FlavorDB (Garg et al., 2017)	FlavorDB contains 25,595 flavor molecules and 1101 receptors	https://cosylab.iiitd.edu.in/flavordb/
FlavorDB2 (Goel et al., 2024)	Compared to FlavorDB, FlavorDB2 adds more functionality, including chemical properties, regulatory status, consumption statistics, taste or aroma thresholds, reported food category uses, and synthesis scenarios	https://cosylab.iiitd.edu.in/flavordb2/
FoodB	FoodB contains 70,926 compounds and 797 foods	https://foodb.ca/
PlantMolecularTasteDB (Gradinaru et al., 2022)	The taste active phytochemical database includes 189 astringent, 1114 bitter, 224 pungent, 263 sweet, and other taste information	https://www.plantmoleculartastedb.org/
<i>ChemTastesDB</i> (Rojas et al., 2022)	<i>ChemTastesDB</i> includes 2944 validated and organized compounds classified into 5 basic taste categories and 4 non-basic taste categories	https://zenodo.org/records/5747393
Flavornet (Arn & Acree, 1998)	Flavornet includes slightly less than 1000 odorants	https://www.flavornet.org/
VCF	Volatile compounds in food, the VCF database contains published information on volatile compounds found in 1445 natural (processed) foods	https://www.vcf-online.nl/VcfHome.cfm
OlfactionBase (Sharma et al., 2021)	OlfactionBase includes 106 main odors, 572 sub-odor types, 3985 odor molecules, 1124 odorless compounds, and 2067 ORs	https://olfab.iiita.ac.in/olfactionbase/
Flavor Ingredient Library	The Flavor Ingredient Library provides information on ingredients that have been determined by an FEMA expert panel that generally considered to be safe under the conditions of their intended use	https://www.femaflavor.org/flavor-library
M2OR (Lalis et al., 2023)	The molecule to olfactory receptor database collects a database of 75,050 bioassay experiments involving 51,395 different OR molecular pairs	https://m2or.chemsensim.fr/
COD (Janfaza et al., 2017)	Cancer Odor Database contains more than 1300 records	http://bioinf.modares.ac.ir/software/cod
ODORactor (Liu et al., 2011)	ODORactor has a total of 3038 odorants and 1608 ORs	https://mdl.shsmu.edu.cn/ODORactor/
HORDE (Olender et al., 2013)	HORDE (Human Olfactory Data Explorer) is a database of human olfactory receptors (ORs), which is the largest multigene family in vertebrates	https://genome.weizmann.ac.il/horde/
mVOC 3.0 (Lemfack et al., 2018)	Microbial volatile organic compound database	https://bioinformatics.charite.de/mvoc/
OVPD (Xue et al., 2022)	EEG database of odor video-evoked physiological signals for emotion recognition, including 32 channels of EEG signals from 10 subjects	https://iiphci.ahu.edu.cn/toxiujue
BIOPEP-UWM Virtual (Minkiewicz et al., 2019)	The sensory peptides and amino acids database of BIOPEP-UWM Virtual contains 582 taste peptides	https://biochemia.uwm.edu.pl/en/biopep-uwm-2/
TastePeptidesDB (Cui, Zhang, Zhou, Zhou, Zhang, et al., 2023)	TPDB records 1950 taste information associated with 943 peptides	http://tastepeptides-meta.com/TastePeptidesDB
Umami DataBase	About 800 items are listed in the Umami Database	https://www.umamiinfo.com/umamidb/
BitterDB (Dagan-Wiener et al., 2018)	Database contains 1041 bitter substances and 72 receptors	https://bitterdb.agri.huji.ac.il/dbbitter.php
SweetenersDB (Chéron et al., 2017)	The database contains more than 316 sweet molecules	http://sebfiorucci.free.fr/SweetenersDB/

(Continues)

TABLE 1 (Continued)

Name	Description (size)	Web site
PungentDB (Chen et al., 2024)	The database documents 205 unique pungent flavor compounds from 231 traditional Chinese medicines, focusing on the compounds' unique pungent flavors—taste, aroma, and thermal sensations—and their interaction with taste and olfactory receptors	http://www.pungentdb.org.cn/home
KokumiPD (He et al., 2024)	The KokumiPD database contains approximately 300 strong-flavor substances	https://www.kokumipd.com/database

Note: (1) Some databases do not have paper support, so there is no citation information supplied. (2) “Not available” indicates that the official website does not provide detailed statistical information. (3) “Collected in SI” means that the relevant data are in the supplementary materials of the paper. (4) Data accessed on January 20, 2023. (5) Limited by the number of citations, some databases without website servers will be shown in Table S2.

due to their rich flavor and physiological activities. As a database focusing on taste peptides, TastePeptidesDB contains taste information of 1950 peptides from 152 papers, of which bitterness (787 peptides) and umami (509 peptides) are the main tastes represented (Cui, Zhang, Zhou, Zhou, Zhang, et al., 2023). In addition to the taste collection, TastePeptidesDB also comes with corresponding analysis tools, including statistical analysis, batch retrieval, and other special programs. It is worth mentioning that TastePeptidesDB is currently the only database that records the threshold information of taste peptides, providing valuable training data for the regression research of taste peptides. BIOPEP-UWM virtual records the activities of a variety of peptides, including flavor, antioxidant, anti-allergic peptides, and more (Minkiewicz et al., 2019). In terms of single-taste databases, sweetness and bitterness are one of the hot spots in database design. SweetenersDB (Chéron et al., 2017) and SWEET-DB (Loss et al., 2002) are common sweetness databases. BitterDB (Dagan-Wiener et al., 2018) is an earlier proposed bitter substance database, which contains relevant information on more than 500 bitter substances and bitter receptors.

Odors are emotive, irrefutably powerful, and deceptive in nature, and they contribute to our emotional behavior, and influence our mood and thoughts. Odor molecule databases are databases specifically used to store and manage information about molecules related to odors. These databases contain a large amount of detailed data about odor molecules, including their chemical structure, molecular weight (MW), odor characteristics, and olfactory threshold. Researchers, perfume manufacturers, food scientists, and so on can use these databases to research and develop new products and conduct odor analysis and prediction. The establishment of odor molecule databases helps to systematically understand the characteristics of various odor molecules and their interactions and improves the research efficiency of odor science. OlfactionBase includes 106 main odors, 572 sub-odor types, 3985 odor molecules, 1124 odorless compounds, and 2067 ORs, which classifies common flavor substances according

to 9 known aroma-classification systems (wine, perfumes, urban odor, drinking water, waste water, compost, fragrance, city smell, human perception) (Sharma et al., 2021). Both M2OR (Lalis et al., 2023) and ODORactor (Liu et al., 2011) focus on analyzing the interaction rules between odor molecules and odor receptors from the receptor perspective. ODORactor can predict the odor properties of chemical molecules and predict potential ORs in specific species (Liu et al., 2011). M2OR records valuable information about the associated responses, receptors, molecules, and experiments for each odor molecule, about 51,395 of them (Lalis et al., 2023). Odor molecules are also often applied as indicators, such as a web-based database comprising comprehensive information of volatile organic metabolites of cancer (Janfaza et al., 2017). Microbial volatile organic compound (mVOC) 3.0 is based on a wide range of literature searches on mVOCs, recording information on the release of small molecules by plants, fungi, bacteria, invertebrates, and vertebrates that affect neighboring organisms and communities as well as hosts (Lemfack et al., 2018). The VCF database focuses on the composition of volatile components in food ingredients, recording volatile component data for a variety of foods, including wine, spices, fruits, fish, and meat.

Although database research seems to have made many achievements at this stage, there are still many difficulties that need to be noted and overcome: (1) Database maintenance: Many databases are only maintained for a short period of time, and no attention is paid to data maintenance and updating; (2) Database user experience: In addition to the integrity and timeliness of the data, time and efforts are also needed to improve the database query experience, such as setting up a portal website and adding batch search functions; (3) Unified data format and interface: A unified record format facilitates data exchange and enrichment, thereby accelerating similar research. Therefore, it is recommended to clarify the underlying logic such as service cycle and data interface at the beginning of building the database and to prepare for data updates and service iteration.

Flavor Substances Identification & Prediction

Smart analysis flowchart

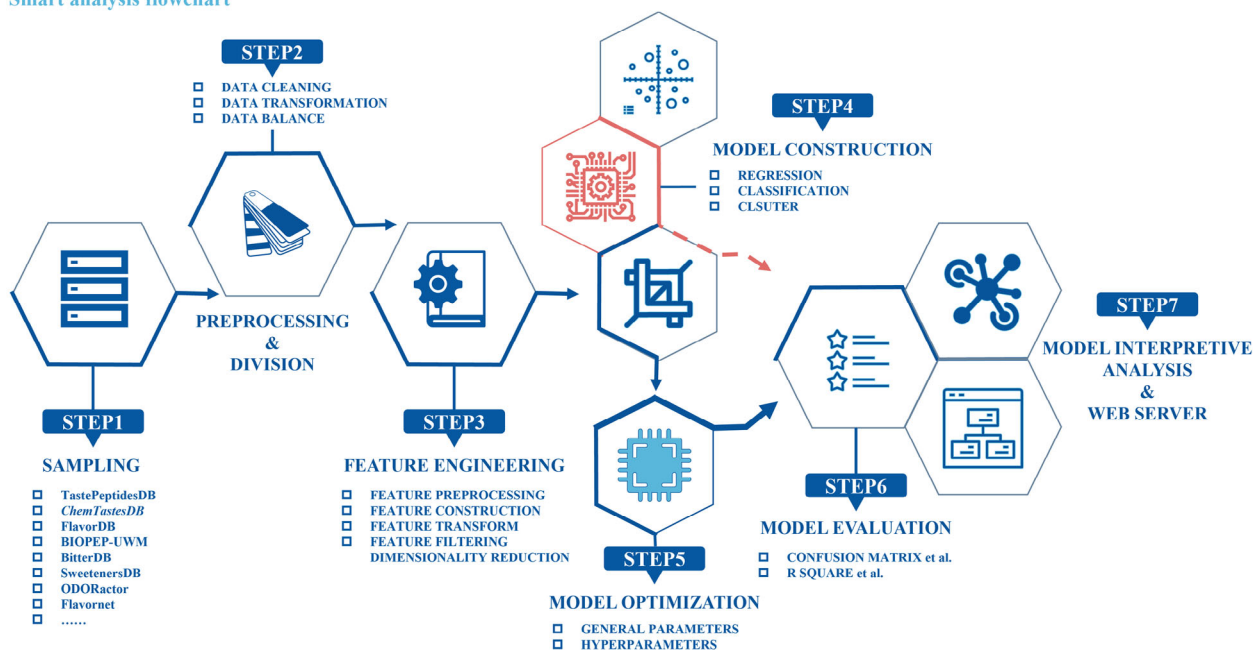


FIGURE 2 Schematic diagram of QSAR modeling steps of flavor substances.

2.4 | Screening model and structure-activity relationship research

The rise of flavor database research has led to a proliferation of algorithms and models for flavor substance prediction (Table 2). Flavor prediction, which involves analyzing and understanding basic senses like odor, sweetness, bitterness, umami, sourness, and saltiness, is an important aspect of molecular research in food chemistry. Through data collection, preprocessing, feature engineering, model construction, optimization, evaluation, and interpretive analysis, ML algorithms learn the characteristics of molecular structures and related flavors in existing datasets to reveal complex recognition patterns (Figure 2). The scorecard evaluation matrix has been shown to be effective in judging bitterness (iBitter-SCM (Charoenkwan, Yana, Schaduengrat, et al., 2020)) and umami (iUmami-SCM (Charoenkwan, Yana, Nantasenamat, et al., 2020)). The judgment model based on characteristic data such as molecular descriptors and molecular fingerprints breaks through the limitations of standard amino acid types and can judge taste peptides and their structural derivatives. Descriptors are a class of methods that characterize molecular structures or certain physicochemical properties through a set of numerical values called molecular fingerprints or descriptors, including molecular properties such as logP, MW, hydrogen bond receptors, and rotatable bonds. According to the difference in the dimensions of describing molecular properties, they are divided into 1D, 2D, 3D, and even 4D (dimensionality based on

molecular dynamics time series) descriptors. The descriptor method has a long history of research on the analysis of small molecule taste activity. In 1966, Deutsch and Hansch established a series of QSAR studies on nine sweet compounds substituted with nitroaniline. They found that the octanol/water partition coefficient and hydrophobicity of the substituents played an important role in the sweetness of these compounds (Deutsch & Hansch, 1966). In 1980–1981, Iwamura used the QSAR model to study the sweetness of 49 perilla, 20 aniline derivatives, and 217 L-aspartyl dipeptide analogs. He proposed that the sweetness of these compounds is highly correlated with the size of the substituents (Iwamura, 1981). In 2006, Tarko used the Preclav software to study the sweetness of aminosuccinic acid derivatives, which found that the sweetness of the compounds was affected by molecular size. The calibration set for this study included 97 molecules ($R^2 = 0.83$), and the validation set included 24 molecules ($R^2 = 0.86$) (Tarko et al., 2006). Yang et al. (2011) established a QSAR model, including 103 sugars and sweeteners, with an R^2 of 0.943 for the test set. In recent years, with the popularization of computer science and modeling algorithms, QSAR is no longer the exclusive property of those who rely on specific software or programming skills. Bioinformaticians can combine more efficient models with professional knowledge backgrounds based on their own understanding to target specific problems. Many studies are devoted to proposing highly specific molecular descriptors. In order to take into account the advantages of multiple algorithms, some groups and algorithms (ensemble learning) are used

TABLE 2 Summary of flavor substances prediction database.

Name	URL	Taste	DataBase	P/N
VirtualTaste (Fritz et al., 2021)	https://insilico-cyp.charite.de/VirtualTaste/	Umami/Bitter/Sweet	SuperSweet/BitterDB/	/
FlavorMiner (Herrera-Rocha et al., 2024)	https://github.com/ipb-halle/FlavorMiner	/	Multiple sources (A)	/
iUmami-SCM (Charoenkwan, Yana, Nantasenamat, et al., 2020)	http://camt.pythonanywhere.com/iUmami-SCM	Umami	UMP442	140/304
UMPred-FRL (Charoenkwan, Nantasenamat, Hasan, et al., 2021)	https://pmlabstack.pythonanywhere.com/UMFRL	Umami	UMP442	140/304
Umami_YYDS (Cui, Zhang, Zhou, Zhou, Zhang, et al., 2023)	http://tastepeptides-meta.com/Umami_YYDS	Umami/Bitter	TPDB-2021	84/149
TPDM (Cui, Zhang, Zhou, Zhou, Meng, et al., 2023)	http://tastepeptides-meta.com/TPDM	Umami/Bitter	TPDB-2022	244/257
Mlp4Umami (M. Liu, Yang, et al., 2024)	https://hwwlab.com/webserver/mlp4umami/predict	Umami	TPDB, UMP	428/299
IUP-BERT (Jiang et al., 2022)	https://www.aibiochem.net/servers/iUP-BERT/	Umami	UMP442	140/304
iBitter-SCM (Charoenkwan, Yana, Schaduengrat, et al., 2020)	http://camt.pythonanywhere.com/iBitter-SCM	Bitter	BIOPEP-BTP640	320/320
iBitter-Fuse (P. Charoenkwan et al., 2021)	http://camt.pythonanywhere.com/iBitter-Fuse	Bitter	BIOPEP-BTP640	320/320
BitterX (Huang et al., 2016)	https://mdl.shsmu.edu.cn/BitterX/	Bitter	BitterDB	260/2379
BERT4Bitter (Charoenkwan, Nantasenamat, Hasan, et al., 2021)	http://pmlab.pythonanywhere.com/BERT4Bit	Bitter	BIOPEP-BTP640	320/320
e-Bitter (Zheng et al., 2018)	https://mdl.shsmu.edu.cn/BitterX/	Bitter	BitterDB/SuperSweet	707/592
BitterSweetForest/SuperSweet (Ahmed et al., 2010)	https://insilico-cyp.charite.de/VirtualTaste/	Bitter/Sweet	BitterDB/SuperSweet	517/685
Sweetpred (Goel et al., 2023)	https://cosylab.iiitd.edu.in/sweetpred/	Sweet	/	811
BitterSweet (Tuwani et al., 2019)	https://github.com/cosylabiiit/bittersweet	Bitter/Sweet	BitterDB/SuperSweet	685/517
Predict Sweetness QSAR (Rojas et al., 2017)	https://micchem.unimib.it/download/matlab-toolboxes/	Sweet	/	435/214
Predisweet (Bouysset et al., 2020)	http://chemosimserver.unice.fr/predisweet/	Sweet	SweetenersDB, SuperNatural II	4796
VirtuousSweetBitter (Maroni et al., 2022)	https://github.com/gabrig88/VirtuousSweetBitter	Sweet/Bitter	SuperSweet et al.	1764/1366
ChemSweet (Qian et al., 2025)	http://chemsweet.ddai.tech	Sweet	/	/
KokumiPD (He et al., 2024)	https://www.kokumipd.com/predict	Kokumi	KokumiPD	300/3962
OdoriFy (Gupta et al., 2021)	https://odorify.ahujalab.iiitd.edu.in/olfy/	Odor	PubChem, the good scents company	679/5474
ODORactor (Liu et al., 2011)	https://mdl.shsmu.edu.cn/ODORactor/	Odor	ODORactorDB	3038/1608
OlfactorySpace (Mayhew et al., 2022)	https://github.com/emayhew/OlfactorySpace	Odor	OlfactorySpaceDB	127
BitterSweet (Tuwani et al., 2019)	https://github.com/cosylabiiit/bittersweet	Bitter/Sweet	BitterDB/SuperSweet	685/517

(Continues)

TABLE 2 (Continued)

Name	URL	Taste	DataBase	P/N
BitterPredict (Dagan-Wiener et al., 2017)	https://github.com/Niv-Lab/BitterPredict1	Bitter	BitterDB	687/1573
Bitter-RF (Y.-F. Zhang, Wang, et al., 2023)	https://github.com/ZhangYufei01/Bitter-RF	Bitter	BIOPEP-BTP640	320/320

Note: (1) The data collected for FlavorMiner were from multiple databases, including FlavorDB, Sigma-Aldrich catalog, BitterSweet, ChemTastesDB, Flavornet, Off-flavor, AromaDB, OlfactionBase, and Natural_tAS2r_agonists. (2) Limited by the number of citations, some serves without website servers will be shown in Table S3.

by scholars. e-Sweet uses a regression model to fit 530 sweeteners and 850 non-sweeteners, achieving an accuracy of 0.91 ± 0.01 (Zheng et al., 2019). BERT4Bitter uses BTP640 as a benchmark dataset (consisting of a set of 320 non-redundant bitter peptides and 320 non-bitter peptides, sorted out in iBitter-SCM (Charoenkwan, Yana, Schaduan-grat, et al., 2020)) and regards each peptide sequence as a corpus based on natural language processing technology, where each of the 20 amino acids is regarded as a word, and bidirectional encoder representation from transformers (BERT) is used for modeling, achieving a test set accuracy of 0.922 (Charoenkwan, Nantasenamat, Hasan et al., 2021). VirtualTaste integrates previously published taste judgment models trained with datasets, including SuperSweet (the total number of compounds in the training set is 1608 molecules, and the test set contains 403 molecules), BitterDB (the total number of compounds in the training set is 1289 molecules), and the sour data (this dataset consists of 1214 training set molecules and 133 test set molecules), achieving high accuracy (all greater than 0.89) in the prediction of sweetness, bitterness, and sourness (Fritz et al., 2021). Bo et al. (2022) used a multi-layer perceptron to process molecular descriptors and molecular fingerprint data and constructed an MLP-Descriptor/FingerPrint prediction model, achieving an AUC value of 0.94.

Feature data can be matched not only with a single algorithm but also with multiple algorithms. Scholars call this idea an integrated model. Common methods can be divided into voting and fitting methods. As the name suggests, the voting method is to make a comprehensive vote based on the results of multiple sub-models. e-Bitter uses a consensus voting strategy of multiple ML methods and achieved an accuracy of 0.929 under the training of 707 bittering agents and 592 non-bittering agents (Zheng et al., 2018). UMPred-FRL is an ML meta-predictor of umami taste that combines six well-known ML algorithms (extreme randomized trees, k-NN, logistic regression, partial least squares, RF, and SVM) and seven features (amino acid composition, amphiphilic pseudo-amino acid composition, dipeptide composition, composition, transition, distribution, and pseudo-amino acid composition) to achieve an accuracy of 0.860 (Charoenkwan, Nantasenamat, Hasan et al., 2021). Similarly, TPDM achieves the

simultaneous judgment of dual taste peptides of bitterness and umami through its TPDM-Bitter and TPDM-Umami sub-models, achieving an accuracy of 0.90–0.94 in 31 bitter/umami peptides (Cui, Zhang, Zhou, Zhou, Meng, et al., 2023). Compared with taste substances, the diversity and complexity of odorants make model research challenging.

Accurate odors prediction is still difficult due to the large-scale chemoinformatics characteristics of odor molecules. OdoriFy can classify user-provided chemicals into olfactory/non-olfactory substances and identify ORs that respond to the query olfactory odor and then perform validation analysis of odor-OR interactions (Gupta et al., 2021). ODORactor is another open-access web server for identifying ORs for small molecules and browsing existing OR-ligand pairs (Liu et al., 2011). A major challenge is that the perceptual qualities vary widely among individuals, which is due to their different genetic and cultural backgrounds. Furthermore, the combinatorial interactions among multiple odorant receptors and different molecules add complexity to the olfactory predictions. At present, most research can only succeed in predicting a limited number of odor descriptions. For example, the famous DREAM challenge predicts 19 rated semantic descriptors, including “garlic,” “fish,” “sweet,” “fruit,” “burnt,” “spices,” “flower,” and “sour” (Keller et al., 2017). Although POM can predict odor intensity and perceptual similarity among 400 out-of-sample odorants (Lee et al., 2023), there could be 500,000 potential olfactory molecules distributed within 150-dimensional perceptual spaces (Kowalewski et al., 2021), or even 40 billion species (Mayhew et al., 2022). Some chemical criteria for odorants are proposed in OlfactorySpace: (1) They are volatile enough to enter the air phase; (2) they are non-volatile but sufficiently hydrophilic enough to be adsorbed into the mucosal layer of the olfactory epithelium; (3) they are hydrophobic enough to bind to the OR pocket; and (4) they can activate at least one OR (Mayhew et al., 2022). Salles et al. suggested that aroma compounds are first absorbed by the oral mucosa and then released into the oral cavity after causing equilibrium disruption during oral swallowing. It has been observed that hydrophilic compounds generally maintain odor for a longer period of time,

whereas hydrophobic compounds are less time-persistent (Salles et al., 2010).

In addition to optimizing the judgment results of multiple models, the integrated model can also enrich multiple different types of sub-models to build a multifunctional platform, greatly shortening the entire identification cycle and improving accuracy. FlavorMiner (currently in preprint status) is a flavor screening platform that first matches the SMILES file uploaded by the visitor with the existing database and directly calls the result if the match is successful (Herrera-Rocha et al., 2024). If no reported information is found, a random forest model, including bitterness, sourness, and fruitiness, is called for prediction. ChemSweet is an AI platform for rapid discovery of potential sweet molecules. It achieves the prediction of physicochemical properties, sweetness characteristics, and health risks by establishing ML prediction models of multiple physicochemical properties and toxicity characteristics, respectively (Qian et al., 2025).

Through model interpretation analysis, we can try to crack the judgment logic behind the black box model or build a simple model to help scientists understand the structural laws behind flavor molecules. A model interpreter algorithm based on game theory, SHAP, has achieved good application in models such as random forests (Lundberg et al., 2020). The TPDM model discovered some key characteristic substructures through SHAP, such as Morgan fingerprints No. 598, 322, 952, 1907, and 1150 (Cui, Zhang, Zhou, Zhou, Meng, et al., 2023). These fingerprints answer the potential taste structures in umami/bitter molecules from a structural perspective. UMPred-FRL uses the SHAP algorithm to rank the importance of sub-models and finds that PLS-AAC, SVM-AAC, and SVM-CTDC models are more important, indicating the importance of PLS-AAC and SVM-AAC features (Charoenkwan, Nantasenamat, Hasan et al., 2021). In the study of taste peptides, the amino acid sequence (primary structure) and spatial structure of peptide active substances are the key to taste receptor recognition (Liu et al., 2019). Some changes in the chemical structure of a substance can change a sweet taste to a bitter or tasteless taste, indicating that small flavor molecules also have an activity cliff phenomenon. The sweetener saccharin is made bitter by the connection of a nitro group on carbon 5 (5-nitroso saccharin), whereas the connection of this group on carbon 4 produces a sweet/bitter tastant (*p*-nitroso saccharin). Interestingly, some tasteless compounds, such as miraculin and cyclin, act as taste modulators, and specifically, these compounds can change the sour taste in a substance to a sweet taste. On the other hand, the presence of an amino group produces a sweet/tasteless compound (6-aminosaccharin) or a tasteless molecule (5-aminosaccharin) (Rojas et al., 2022). Despite the great

progress in the study of structure–activity relationships, it is difficult to establish a universal theory between molecular structure and sensory quality, because the system established by humans is time-consuming and imperfect (Ji et al., 2023). Fortunately, uncovering new flavor scaffolds can significantly enrich the flavor family and accelerate QSAR research studies (Bouysset et al., 2020).

3 | FLAVOR SUBSTANCE GENERATION UNDER THE GUIDANCE OF AI

Compared with mechanism-based QSAR research, end-to-end learning technologies like deep learning that do not require feature engineering have been widely used in recent years, especially for molecular generation technologies such as CharRNN, REINVENT, AAE, VAE, ORGAN, and LatentGAN and GraphINVENT (Kou et al., 2023). These technologies differ from traditional flavor substance design in that they construct functions mapping desired attributes or characteristics to a multidimensional space. This allows the molecule generation model to quickly, accurately, and reproducibly generate molecules with the desired flavor properties. The ability and applicability of the generated model mainly depend on the quality of the training data (its diversity and representativeness) and the architecture of the model (Kou et al., 2023). Inverse design methods aim to optimize the corresponding chemical structure starting from the desired properties. From an ML perspective, the inverse design problem can be solved by so-called generative modeling. Mathematically, the model is defined by learning the probability distribution function of the properties of a given molecule. In contrast, generative models aim to exploit the joint probability of chemical species with target properties. The overall idea of generative modeling is to implement a system that is able to generate new compounds that are expected to have a set of desired chemical features, thereby effectively circumventing the problems in the forward design process. Generative model methods are diverse and have different internal mechanisms. Despite this diversity, the ultimate goal is shared: to explore unknown areas of the chemical design space and find high-performance molecules that can be synthesized and meet the requirements (Ansine & Isayev, 2023). MolFilterGAN was proposed to use adversarial networks to generate molecules that were more like known drugs or bioactive molecules and to produce more diverse and balanced negative samples through a progressive enhancement strategy to improve the discrimination and generalization capabilities of the discriminator (X. Liu, Zhang, et al., 2023). Fragment growing is also an effective ligand design strategy. FragGrow allows users to set a series of constraints when searching for the desired

fragment, including the ability of the fragment to interact with specific protein sites and then generate new molecules by directly replacing hydrogen atoms or substructures or by virtual synthesis (Zhang et al., 2024). As a general framework dedicated to the design of small molecule antibiotics, AutoMolDesigner contains two functional modules, namely, molecular generation that supports deep learning and automated antimicrobial activity/property prediction based on ML (Shen et al., 2024). In addition, AutoMolDesigner has designed an intuitive graphical user interface and a corresponding command line interface for Windows and MacOS platforms. The software has also been tested on personal computers with various device configurations. It can run on laptops without independent graphics processing units and achieve performance close to that of high-performance servers. To date, molecule generation algorithms have been extensively used in new drug discovery, whereas their application in the food field has been less prevalent. Luana P. Queiroz et al. applied a novel deep learning framework SciML, where molecules are designed through random sampling actions, leading to the discovery of potential flavors like hinokitiol and methyprylon (Queiroz et al., 2023a). A combined system consisting of generative and reinforcement learning models was proposed to generate flavor molecules that could obtain better synthetic accessibility scores and natural product similarity scores (Queiroz et al., 2023b). A transfer learning method was then added to the generative model to form a flavor engineering framework, which is divided into three layers: The first, second, and third were the generative model, the reinforcement learning model, and the transfer learning model. Through this framework, molecules with specific flavor descriptors could be directly designed and obtained, including sweet, fruity, green, woody, fresh, spicy, citrus, fatty, and ripe molecules (Queiroz et al., 2023c). In addition, protein design provides new protein sources and technical support for protein transition. By using computational methods and bioengineering techniques, scientists can develop proteins with specific functions and properties, such as unique flavors, altered textures, improved nutritional value, or flavored proteases that can be cleaved to produce specific nutrients and functions. Protein design based on specific nutritional and functional activity motifs contributes significantly to food sustainability (Notin et al., 2024).

4 | NEW IDEAS FOR PRODUCT DESIGN IN THE ERA OF AI

Global food systems are facing the challenge of providing healthy and adequate food in a sustainable way, which is exacerbated by the impacts of climate change and the

increasing protein demands from a growing world population. Fortunately, by optimizing diets or developing new food products, this challenge can be addressed without compromising nutritional level, thereby improving the sustainability of our food systems.

4.1 | Food pairing, food-drug interactions guided by big data

The theory of food pairing was first proposed in 2002 by Michelin-starred chef Heston Blumenthal and flavor expert François Benzi (of Firmenich in Geneva), and the premise is simple: The more aroma compounds two foods have in common, the better they taste together. It is based on the fact that flavor is determined primarily by volatile aroma compounds, rather than by mouthfeel and taste (de Klepper, 2011). A classic example is chocolate and blue cheese, which share at least 73 compounds named (Galmarini, 2020). Computational gastronomy can visualize possible ingredient combinations based on volatile components through flavor network graphs. Ahn et al. constructed a network-based approach to associate ingredients (381) used in recipes around the world with the compounds (1021) that contribute to the flavor of each ingredient. Their study found that North American and Western European dishes tend to combine ingredients that share flavor compounds, whereas East Asian dishes avoid them. However, a shortcoming of their study is that the method did not measure the concentration of the compounds with a threshold (Ahn et al., 2011). Arellano et al. extracted flavor pairings of beer from consumers in four different countries (Argentina, Colombia, Peru, and Mexico) from social media. They constructed a similarity matrix using the co-occurrence of flavor pairings in social media and analyzed it using multidimensional scaling to find the pairing patterns for each country. They obtained a map that helps understand the cultural differences in flavor pairings for each country (Arellano-Covarrubias et al., 2019). Some studies have found a high co-occurrence rate between beer and wine. This may not mean that wine and beer go well together, but rather that the process of word association links the two alcoholic beverages (Guerrero et al., 2010). Unlike talented chefs who rely on their intuition to pair foods, FlavorGraph is a network of foods and compounds that uses metapath2vec for scientific knowledge association and statistical analysis to provide 15 effective recommendations for food pairing and predict the relationship between foods (Park et al., 2021). WineGraph, an extended version of FlavorGraph, uses a food dataset containing 500,000 reviews and a wine review dataset containing more than 130,000 entries to pair food and wine based on taste and rules defined by sommeliers (Gawrysiak

et al., 2024). It is important to note that there are individual differences in the perception of food pairing, which emphasizes the importance of liking (liking each product, tasted separately). This is undoubtedly a significant issue. However, other individual differences based on attitudes and motivations, such as health concerns, cultural characteristics, or social influences, may be equally important as they influence eating behavior (Eschevins et al., 2019).

Generally Recognized As Safe (GRAS) chemicals and inactive ingredients (IIGs) were previously considered reliable sources for drug delivery, formulation science, and food production. However, an increasing number of studies and clinical reports have begun to question their bioinert properties (Reker et al., 2019). Some studies have demonstrated that certain foods can affect the activity of various drugs by increasing drug's metabolism, reducing drug's bioavailability, or responding to adverse drug events (ADEs), and these phenomena are called food-drug interactions (FDIs) (Tseng et al., 2023). FDMine was proposed to discover new FDIs and examine possible pharmacological effects of food compounds. It extracted data from the FoodDB and DrugBank databases to construct networks based on structural similarity (food-drug similarity, drug-drug similarity, and food-food similarity networks), which have been found to explain the anti-inflammatory effects of foods (Rahman et al., 2022). An ML framework was proposed to identify potential active species of *P*-glycoprotein (*P*-gp) and uridine diphosphate glucuronosyltransferase-2B7 (UGT2B7) from GRAS and IIGs. Vitamin A palmitate and abietic acid were successfully identified and verified through its *in vivo* experiments (Reker et al., 2020). One study used ML techniques to predict food effects from an extensive database containing more than 300 drugs with multiple chemical profiles and more than 20 drug properties or characteristics. Through model interpretive analysis, it was found that cDose number (minVSLgS 3–7.5 and ALOGPS-based), an indicator of the solubility–dose relationship, was important (Gavins et al., 2022). To bridge the gap between the food and biomedical fields, some studies focused on extracting or summarizing FDI effects from existing data/experiments with the help of LLMs. A three-round iterative prompting method for food effect summarization was proposed to prove that LLMs such as ChatGPT and GPT-4 could effectively assist humans in summarizing food effects from new drug applications (Shi et al., 2023). It has been demonstrated that three natural language processing pipelines (FoodDis, FoodChem, and ChemDis) for relationship mining that generate knowledge graphs constructed from triplets in the form of (entity1, relationship, entity2) could extract relationships among food, chemical, and disease entities from the raw text of biomedical text data summaries with an average accuracy of about 70% (Cenikj et al., 2023). FoodGPT

introduced incremental pre-training for structured knowledge and scientific document processing and reduced the problem of machine hallucination (Qi et al., 2023).

4.2 | Personalized dietary and complementary medical advice based on multimodal omics data

The development of advanced computers and mathematical algorithms has enabled the capability of processing large amounts of data and made it possible to use omics technology (transcriptomics, metabolomics, neuroomics, microbiomics, genomics, proteomics, epigenomics) to enhance human health (Figure 3a). By integrating internet data to collect knowledge related to diseases, medicines, food, nutrition, and so on, named entity recognition and named entity linking are associated to build a knowledge graph to provide suggestions and guidance for designing products that take into account nutrition, health, diet, sensory, physical, psychological, and other factors (Figure 3b). One of the major challenges in clinical nutrition today is precision nutrition, which aims to translate the increasing findings from basic nutritional science into meaningful and clinically relevant dietary recommendations (De Toro-Martin et al., 2017). A study by David Zeevi et al. continuously monitored the post-meal blood sugar levels of 800 people for a week. They found significant differences in how individuals responded to identical meals, suggesting that universal dietary recommendations might not be universally applicable. They designed an ML algorithm that integrated various factors such as blood parameters, dietary habits, anthropometry, physical activity, and gut microbiota, which could accurately predict personalized postprandial glycemic responses to real-life meals (Zeevi et al., 2015). Figure 3c fully illustrates the integration of personal information (food intake, clinical knowledge, and omics data) and information-based taste research to improve food quality, taking flavor and nutrition into account. Technology for digital characterization of food components has made great progress in recent years. MIT Dagdeviren's team designed a flexible sensor that can be rolled up and swallowed. The sensor adheres to the stomach or intestinal wall and measures food in the stomach, helping doctors monitor the food intake of patients being treated for obesity (Dagdeviren et al., 2017). In addition to device detection, program development based on users' portable smart devices has also achieved good results. Amazon's Calorie Mama can quickly identify food and protein content by searching and training in global food libraries. There was also the Im2Calories system developed by Google, which has achieved good results in combining visual analysis, image recognition, and large

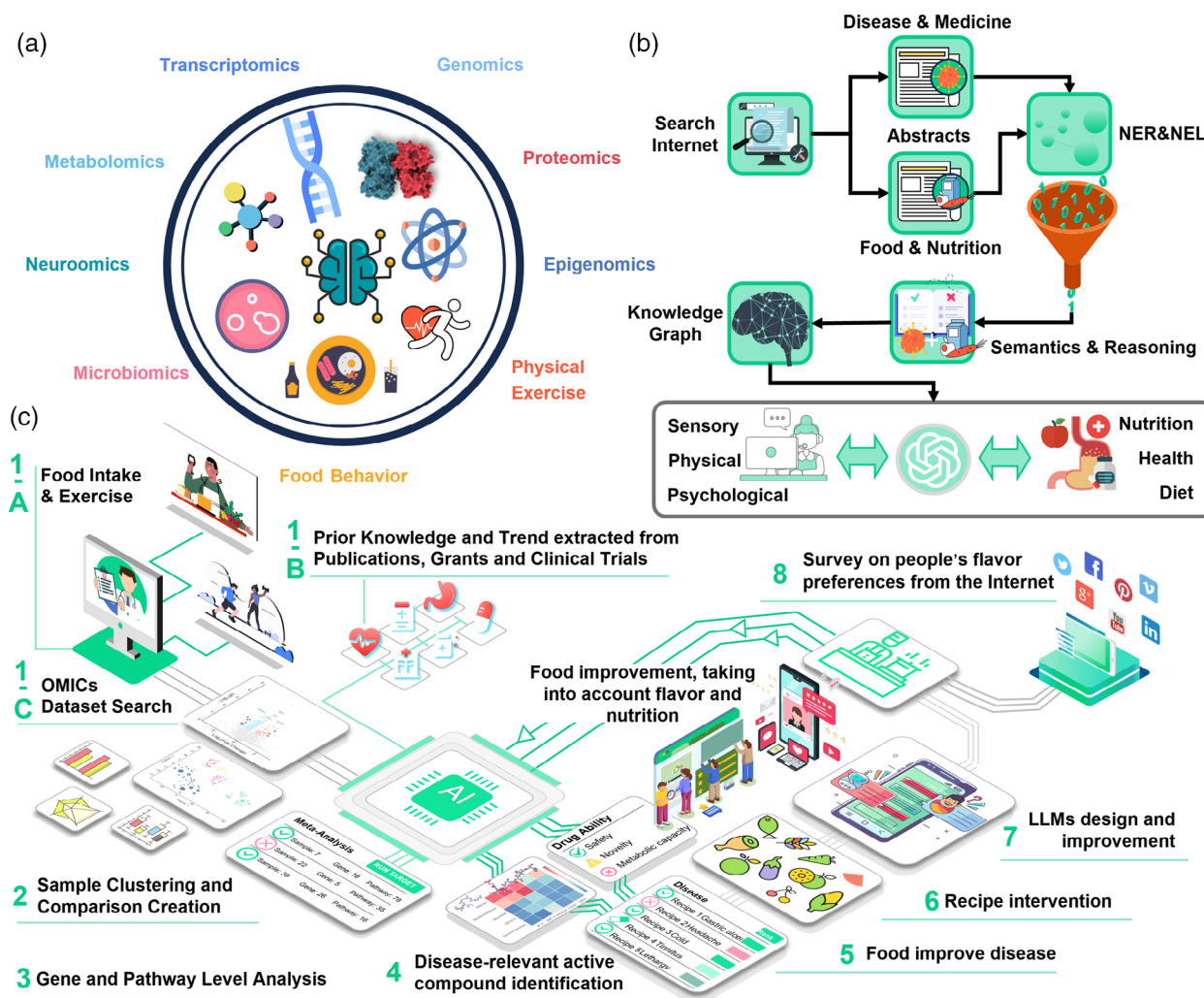


FIGURE 3 (a) Display of available data for multi-omics data and individual information. (b) Flowchart of extracting professional knowledge through large language models (LLMs) to mine the relationships among food, medicine, and diseases, and providing personalized dietary advice through the mined relationships combined with artificial intelligence (AI). Among them, NER (named entity recognition) and NEL (named entity linking) are inherent operations in natural language processing, which, respectively, link the entities mentioned in the text to the corresponding entities in the knowledge base and identify specific types of entities in the text and sort them into predefined categories. (c) Schematic diagram of the method of providing personalized dietary guidance based on the fusion of food intake, physical exercise, advanced medical knowledge, and omics data, and improving the method based on personal taste.

databases to confirm food origins, food classification, and calorie prediction (Myers et al., 2015). UMDFood-VL was proposed to combine visual language models (VLMs) with chemical analysis technology. Using the huge UMDFood-90k database, an advanced VLM was introduced, which greatly improved the speed and accuracy of the nutrient estimation process. The macro AUCROC of this model for lipid quantification was 0.921 (Ma et al., 2024).

With the development of food chemistry and nutritional science, food ingredients and the relationship between diet and health have received more and more attention. The Healthy Diets Monitoring Initiative is a joint effort of the UN Food and Agriculture Organization, the United Nations Children's Fund, and the World Health Organi-

zation to translate scientific consensus into national and global healthy diet monitoring guidelines (Neufeld et al., 2024). It is important to emphasize nutritional guidance and educational programs on healthy dietary choices. Utilizing food composition data to develop therapeutic diets to address chronic diseases like obesity, cardiovascular diseases, and food allergies has become clinically feasible (Tseng et al., 2023). O'Donovan et al. applied a hybrid dietary model, a physiology-based computational model, to simulate an individual's metabolic health. The model can also quantify liver fat accumulation and beta cell function and use meal response data to generate a personalized dietary model (O'Donovan et al., 2024). At this stage, relevant business activities have been carried out and

attempted. Companies, like Campbell's Soup and Nestlé, have started personalized nutrition services using tests that cover 60 physiological indicators, DNA and blood tests, and other components. Precision nutrition and personalized customization technologies based on multi-omics are accelerating the birth of new food formats. In addition, comprehensive and transparent nutrition labels enable consumers to make informed choices among similar products. Food Compass incorporates a wider range of food characteristics, attributes, and unified scoring principles. It scores 54 attributes across 9 health-related areas: nutrient proportions, vitamins, minerals, food ingredients, additives, processing, specific lipids, fiber and protein, and phytochemicals. This information, presented via front-of-pack labels, warning labels, taxes, company ratings, and so on, helps differentiate the healthfulness of foods, which can guide consumer choices, research, food policy, industry reformulation, and mission-focused investment decisions (Mozaffarian et al., 2021). FoodSky is a Chinese language model for the Chinese food industry. It can realize a series of tasks in the food industry, including recipe generation, food calorie calculation, intelligent nutritional meal matching, personalized dietary recommendations for different groups of people, and diet-disease association mining (Zhou et al., 2024).

In general, diet is a modifiable risk factor that plays a key role in various chronic diseases and even most diseases such as cancer (Adolph & Tilg, 2024). However, our understanding of the mechanism of action of chemical compounds in food is still incomplete. On the one hand, due to the complexity of each trace component in food, the identification of this “dark matter” is challenging in itself (Barabási et al., 2020); on the other hand, how to use AI to extract patterns about food components, processing, and flavors from massive dietary exposure records to build a molecular network linking diet and health requires continuous practice and exploration (Menichetti et al., 2024). The advent of the AI algorithm and big data era has brought hope for identifying complex and multi-level relationships such as food, dietary behavior, metabolic regulation, energy balance, and flavor improvement, dietary pairing, dietary records, nutritional recommendations, and health monitoring (Liang et al., 2024).

4.3 | Food formulation enhancement and improvement for balancing nutrition and flavor

To transform food systems, it is essential to comprehensively understand daily food preparation and consumption practices, including their impacts on health and sustainability. Deep generative models are effective tools in the

reverse design process, as they can effectively learn product information and propose new product candidates (Al-Sarayreh et al., 2023). The food supply is growing at an alarming rate, and the composition of agricultural and processed foods is changing. In addition, scientific understanding of the relationship between dietary intake and health is accelerating new demands for knowledge about foods and their composition. These demands increase the need for transparent and easily accessible information about nutrients and other food ingredients. Table 3 summarizes the common, publicly accessible, mainstream food ingredients and recipe databases that provide data support for AI research. The complex ingredients and various compounds in food can all be used to construct information-rich food databases. FoodData Central was proposed to collate the US Department of Agriculture's food composition data, including nutrient content and other food composition values for a variety of basic foods (unprocessed or minimally processed foods), and provide a large amount of basic metadata, including sample size, sampling location, collection date, analytical method used, and (if applicable) agricultural information. The international network of food data systems was established in 1984. It is a global network of food composition experts that aims to improve the quality, availability, reliability, and use of food composition data, providing guidelines, standards, compilation tools, databases, capacity development tools, policy recommendations, advocacy tools, and technical assistance. Some databases that focus on recording national food data at the national level can better serve the people of various countries, such as the database (NZFCD) that collects about 2700 New Zealand foods, the AFCD that focuses on food safety standards in Australia and New Zealand, and the Malaysian database (MyFod) that collects food composition and nutrients of 441 raw and processed food materials, and 668 prepared foods. In addition, FooDB 1.0 is the world's largest and most comprehensive resource for food composition, chemistry, and biology. It provides information on macronutrients and micronutrients, including many of the components that give foods flavor, color, taste, texture, and aroma. Users can browse or search FooDB by food origin, name, descriptor, function, or concentration. Depending on personal preference, users can view FooDB's contents via Food Browse (listing foods by chemical composition) or Compound Browse (listing chemicals by food origin). Phenol-Explorer is the first comprehensive database on the polyphenol content of foods. The database contains more than 35,000 values for 500 different polyphenols in more than 400 foods. It is continuously maintained, with comprehensive data on polyphenol metabolism added in Phenol-Explorer 2.0 and data on the effects of food processing and cooking introduced in Phenol-Explorer 3.0. AdditiveChem

TABLE 3 Publicly accessible online databases of mainstream food ingredients and recipes.

Name	Descriptor	Website
FoodData Central (USDA)	The database provides food nutrient data, ingredients, and metadata of 35,927 types of foods	https://fdc.nal.usda.gov/
EuroFIR	Food data are the core of the work of EuroFIR (European Food Information Resource) that contains food composition information	https://www.eurofir.org/food-information/
InFoods	Provide information on food ingredients, nutrition labels, supplements, and biodiversity data to meet the needs of different users	https://www.fao.org/infoods/infoods/en/
AdditiveChem	The database contains more than 9064 food additives, their molecular structures, chemical and physical properties, absorption, distribution, metabolism, excretion and toxicity properties, biosynthetic and biodegradation methods, usage specifications, toxicological and risk assessment data, and human targets extracted from 16 databases	http://www.rxnfinder.org/additivechem/
SuperNatural III	SuperNatural 3.0 contains 790,096 different structures (including isomers) and 449,058 unique natural compounds with their structural and physicochemical information	https://bioinf-applied.charite.de/supernatural_3/index.php
PhytoHub	PhytoHub contains detailed information on dietary phytochemicals and their human and animal metabolites. The information of human or animal metabolites, chemical structures and identifiers, major dietary sources, monoisotopic masses, and spectral data are also collected	http://phytohub.eu/
Phenol-Explorer	Phenol-Explorer is the first comprehensive database on polyphenol content in foods, containing more than 35,000 content values for 500 different polyphenols in more than 400 foods	http://phenol-explorer.eu/
FoodDB	FoodDB contains 70,926 compounds and 797 foods (accessed on January 20, 2023)	https://foodb.ca/
NZFCD	It contains the nutritional contents of over 2700 foods commonly prepared and eaten in New Zealand	https://www.foodcomposition.co.nz/
AFCD	The database provides nutritional data for 1616 common foods available in Australia and up to 256 nutrients for each food	https://www.foodstandards.gov.au/
Malaysian Food Composition Database (MYFCD)	The database provides food composition and nutrients of 441 raw and processed food materials, and 668 prepared foods	https://myfcd.moh.gov.my/
Dutch Food Composition Database	The database provides the composition and 130 nutrients of 2207 commonly eaten foods by a large part of the Dutch population	https://www.rivm.nl/en/dutch-food-composition-database
Nutritionix	Verified nutrition database, including 80,5075 items from 39,906 grocery brands, 172,903 items from 799 restaurants, and 13,521 common foods	https://www.nutritionix.com/
CalorieKing	The database provides nutrition facts for branded and fast foods	https://www.calorieking.com/
Food Additive database	The database provides information about the food additives approved for use in food in the EU and their conditions of use	https://webgate.ec.europa.eu/
My Food Data	Includes nutritional facts for over 600,000 foods	https://www.myfooddata.com/
FoodOn	FoodOn is an ontology, a controlled vocabulary available to humans and computers that focuses on statements and naming of foods that can be entered into the database to query or reason about	https://foodon.org/
Allrecipes	A website provides ingredients, cooking directions, nutritional facts, images, user reviews, and ratings	https://www.allrecipes.com/recipes/
Cookpad	A Japanese cooking website provides ingredients, cooking directions, nutritional facts, images, user reviews, and ratings	https://cookpad.com/

(Continues)

TABLE 3 (Continued)

Name	Descriptor	Website
Xiachufang	A Chinese cooking website provides ingredients, cooking directions, nutritional facts, images, and user reviews. Information can be searched by ingredients and dining occasions	https://www.xiachufang.com/
Epic Kitchen	A website provides egocentric, multi-faceted, audiovisual, non-scripted home cooking recordings	https://epic-kitchens.github.io/2021
BBC goodfood	A website provides ingredients, cooking directions, nutritional facts, ratings, and dish pairing recommendations	https://www.bbcgoodfood.com/recipes/
Tasty	A website provides ingredients, cooking directions, and user reviews that can be searched by ingredients	https://tasty.co/
Green Kitchen Stories	A cooking website focuses on vegetarian foods and provides ingredients, cooking directions, cooking stories, closeup featured food photos	https://greenkitchenstories.com/
Love and Lemons	A website provides ingredients and cooking instructions	https://www.loveandlemons.com/
food52	A website provides ingredients, cooking instructions, author notes, and test kitchen notes	https://food52.com/recipes
Kochbar	A website provides ingredients, nutrition profiles, cooking instructions, and consumer ratings	https://www.kochbar.de/
First Foods	Database for baby food	https://solidstarts.com/foods/
FADB (Ginex et al., 2014)	A food additive molecular database	pietrocozzini@unipr.it

contains more than 9064 food additives, as well as their molecular structures, chemical and physical properties, absorption, distribution, metabolism, excretion, and toxicity properties, biosynthesis and biodegradation methods, usage specifications, toxicology and risk assessment data, and targets in the human body (Zhang et al., 2020). The currently available comprehensive food composition databases only cover a small part of the total amount of chemicals present in our food. With the emergence of novel flavor mining technologies and the improvement of detector sensitivity, more and more food components are being reported. Through the scientific literature information mining technology in LLMs, it is possible to build a real-time updated and more comprehensive food composition database. FoodMine aimed to use information mined from research literature to expand the currently available food chemical composition information. Taking garlic and cocoa as an example, the results mined by FoodMine were 137% more than the content of FooDB and USDA databases (Hooton et al., 2020).

The physical and chemical properties of food ingredients are quite complex, and there is a lack of accurate information about food ingredients, as well as a lack of experience in applying AI to the food field. These difficulties have hindered the migration and application of ML applications or deep learning methods in the food field. Table 4 collects and displays the application of LLMs related to product design in the food industry, including informa-

tion such as product platform, description, and source. Shiru has built an AI platform named Flourish, which contains structural data of nearly 450 million proteins found in nature. After scanning nearly 10,000 recipes, it found a structure that could be combined with oil to produce structured fat. This led to the development of OleoPro, a new plant-based protein and fat ingredient that can reduce up to 90% of saturated fat in plant-based meat while improving its processing properties. Japan's Sapporo Brewery and IBM Japan have collaborated to develop the first sparkling alcoholic beverage using AI. The AI system named N-Wing Star analyzed 170 existing products, evaluated approximately 1200 formulas and more than 700 raw materials, and generated the basic formula for this drink with its recommended raw material combinations and the ratio of each material. In addition to product and recipe optimization, AI also helps predict trends of food and fashion. Kerry Trendspotter also uses ML and AI to predict food trends.

5 | AI-BASED PRODUCT DESIGN IN THE INTERNET ERA

The rise of mobile internet technology has led to the emergence of dark kitchen businesses such as online ordering and food evaluation. LLMs aid in understanding and processing these online evaluations, thereby grasping

TABLE 4 Product development and design involving artificial intelligence.

Product (platform)	Description	Organization/Link
Perfume/ScentElligence	The ScentElligence platform is said to boast a deep learning algorithm that analyzes data and delivers fragrant possibilities by combining sensoriality with performance	Cosmo International Franchises/ https://scentelligenceproject.com
Kerry Trendspotter	Using ML and AI to predict food trends	KERRY/ https://www.kerry.com/about/our-company/expertise/consumer-insights/kerry-trendspotter.html
HELL ENERGY Drink	Every link related to formula design, taste evaluation, safety management, and marketing is carefully crafted by AI systems	HELL ENERGY/ https://www.hellenergy.com/world-sensation-a-i-developed-and-tasted-its-own-energy-drink/
N-Wing Star (Breweries)	During the development process, approximately 1200 types of formulations as well as the data on over 700 raw materials that have been evaluated for about 170 existing products were fed into N-Wing Star	Sapporo Breweries/ https://tequilaraiders.com/new-release/sapporo-and-ibm-unveil-new-alcoholic-beverage-created-by-ai/
Notcheese/Giuseppe	A cheese alternative made by plant-based ingredients that melts, stretches, and tastes the same	NotCo & Kraft Debuts/ https://notco.com/cl/productos/notcheese/notcheese-mantecoso
OleoPro/Flourish	This is a new plant-based protein and fat ingredient that can reduce up to 90% of saturated fat in plant-based meat while improving the processing properties of plant-based meat	Shiru/ https://shiru.com/post/announcing-oleopro-2/
Firmenich	A delicious lightly grilled beef taste for use in plant-based meat alternatives	https://www.firmenich.com/taste-and-beyond/blog-article/worlds-first-ai-created-flavor
AI-FOOD	Launching a solution that helps food retailers contribute to a better food system in two ways: On the one hand, using a recommendation system to provide consumers with personalized information and suggestions to improve their eating habits and choices. On the other hand, it helps chain stores increase consumer engagement and make the brand the first choice of consumers when making purchases	https://www.eitfood.eu/projects/ai-food-ai-based-mobile-app
GenAI	Consumer-centric branding: decisions, campaigns, and products	https://tastewise.io/
Forager/Brightseed	Phytonutrient development platform, which has already analyzed 700,000 compounds for their health properties	https://www.brightseedbio.com/
Ada/PepsiCo	Explore new product ideas and find, categorize, and connect millions of consumer insights	/
ONE/McCormick	ONE is used to season food at home and can learn and predict complementary and alternative recipes, adjust the proportion of raw materials, and meet the novelty of flavor recipes	/
Plant Jammer	Intelligently generate recipes based on user-selected food ingredients	/
NotCo Company	Replacing or supplementing outdated ways of growing, producing, and processing food with sustainable alternatives and revolutionizing the vegan food industry with cutting-edge technologies that can replicate the flavor, texture, and color of traditional foods entirely from plant-based ingredients	/
Climax food	Explore smarter pathways from plant to product to create a new generation of plant-based foods at lower cost for consumers and the environment without sacrificing flavor and texture	https://climax.bio/
Beyond honeyco	Making cooking robots cook with AI-driven technology	/
Food pairing	Flavor intelligence map based on molecular analysis	https://www.foodpairing.com/
Gastrograph	Investigate each product using a predictive flavor profile engine to predict how new food products will perform in different markets	https://www.gastrograph.com/

(Continues)

TABLE 4 (Continued)

Product (platform)	Description	Organization/Link
Flavorwiki	Segmenting the Food Industry on Consumer Preferences (food flavor, texture, aroma, and mouthfeel)	https://flavorwiki.com/
Nuritas	Through cutting edge, AI-precision technology, Nuritas is creating a brand-new category of precision, cell-signaling peptide ingredients from nature	https://www.nuritas.com/
TasteWise	Through big data analysis and Li Oh Ajie emerging area is and provides consultation	https://tastewise.io/
Myrissi	Chemical composition begins to drive the company's creative and product development process, with color being the first step to determine the appropriate olfactory associations	https://www.givaudan.com/fragrance-beauty/myrissi
Spoonshot	Predicting people's taste preferences using food science and rich cuisine data	https://spoonshot.com/

Abbreviations: AI, artificial intelligence; ML, machine learning.

consumer psychology (da Cunha et al., 2024). Recommendation systems use algorithms and models to recommend foods that suit consumers' tastes and needs based on their preferences and similar consumption behaviors, thereby increasing purchase conversion rates and sales. With the help of big data analysis, companies can understand consumers' purchase history, preferences, and behaviors and provide personalized product recommendations, pricing strategies, and promotions, thereby enhancing consumers' purchasing experience and loyalty (Verma et al., 2015). Clootrack, developed by India's Clootrack Software Labs, is an AI-driven data analysis platform. It aims to provide real-time assessment of brand recognition using deep learning-based AI algorithms and reliable mathematical models. A study using ML algorithms to analyze JUUL flavors on Twitter could classify a collection of 30,075 tweets related to JUUL flavors into positive and negative sentiment with 89% accuracy (Malik et al., 2021). ConvBiLSTM, a word embedding model that employs numerical values to represent tweets, is proposed to accurately evaluate the negative perceptions of food delivery services on Swiggy, Zomato, and UberEats and then propose improvements (Vatambeti et al., 2024). Kevin's team conducted extensive chemical analysis and sensory evaluation of 250 different beers, combined with more than 180,000 consumer online evaluation data (from the RateBeer website), and used multiple ML models to successfully build a prediction system for beer flavor and consumer preferences degree model (Schreurs et al., 2024). Although big data technology can promote the development and progress of the food industry, it must be used with caution to avoid negative impacts such as data privacy issues and quality control problems. At the same time, the government should strengthen supervision and legislation to protect the legitimate rights and interests of consumers.

6 | PRECAUTIONS FOR EXPANDING THE APPLICATION OF LLMs IN THE FOOD FIELD

The rapid development of AI has promoted the development of LLMs, with their extraordinary application potential in the food industry extensively examined in the previous chapters of this paper. However, the unrestricted and unscreened use of LLMs can lead to risks and crises, as pointed out below:

1. The safety of AI-generated products is difficult to evaluate: Although the potential benefits of small molecules and protein design far outweigh the risks at this stage, there is still a lack of fast, reasonable, and accurate tools for safety assessment.
2. AI's infringement of intellectual property rights: LLMs' reliance on training data has triggered public concerns about data privacy and security. It is crucial to establish a robust framework to protect sensitive information and prevent unauthorized access or misuse of citizen data. In most cases, operators use LLM-generated results (text or images) to cover up the source of the generated content, potentially leading to infringe intellectual property rights unknowingly (Van Dis et al., 2023). In addition, AI models or algorithms are abstract concepts, which is difficult to obtain patent applications. Therefore, the protection of AI is also a matter that needs to be treated with caution.
3. Scientific integrity and rigor are at risk: LLMs have changed the way scientists find information, conduct research, and write and evaluate publications. The widespread use of commercial "black box" AI tools in research could introduce bias and inaccuracies, undermining the validity of scientific knowledge. The generated output may distort scientific facts but still

sound authoritative. For example, one researcher used the technology behind the ChatGPT to fabricate a clinical trial dataset to support its unverified scientific claims (Taloni et al., 2023).

4. There is hidden bias in AI processing results: Limited by the dataset imbalance and overfitting effects, the AI model can easily connect and deeply bind the non-objective knowledge in the sample. Pratyusha Ria Kalluri, an AI graduate student at Stanford University, and her colleagues found that the images generated by Stable Diffusion, a popular tool released by Stability AI, and OpenAI's DALL-E have overwhelmingly adopted common stereotypes, such as associating Africa with poverty. LLMs, including those that power chatbots like ChatGPT, make racially biased judgments based on the user's dialect (Hofmann et al., 2024).

At this stage, global governments are initiating regulations for AI technology, but comprehensive and effective legislation will take several years. The draft EU AI Bill requires the disclosure of AI-generated content and the release of excerpts of copyright-protected data used to train AI systems. At present, the EU has issued 141 binding policies applicable to AI in the fields of healthcare and population health (Schmidt et al., 2024). China's Cyberspace Administration has announced its intention to enforce AI regulations, including requiring generative AI developers to prevent the spread of misinformation. The UK's government also promotes an intergovernmental agreement to limit AI risks (Bockting et al., 2023).

7 | OUTLOOK

AI can effectively use big data to create customized food options by integrating professional knowledge and actual individual information, thereby achieving precise nutrition for individuals. In recent years, research on taste receptors has progressed rapidly. The perception of taste substances by taste receptors is closely related to the absorption of nutrients and is the basis for the body to ingest, digest, absorb, store, and consume nutrients. Effective ligands for taste receptors have also been used in clinical practice. Taste receptors not only allow people to "feel" the taste of various substances in food but also regulate a variety of physiological activities in the human body. However, the research on taste receptors faces challenges. First, the exact structure of taste receptors has not yet been solved, making it difficult to directly determine the action sites of taste molecules on their receptors, and as well as their conformational changes. However, the effectiveness of LLMs on specific laboratory datasets still needs to be tested in a broad population. Moreover, a larger

and more open comprehensive database needs to be established to collect and utilize data from multiple perspectives such as food and nutrition. Progress in regulatory policies and public perception of LLMs is currently slow and insufficient.

Although most AI applications at this stage are targeted toward automation and drug development, considering the impacts of nutrition and population size, applying this technology to food could benefit a large population and the environment. LLMs with enhanced nutritional counseling can provide sustainable dietary advice, which could reduce the incidence of foodborne diseases due to excessive intake and nutritional imbalance, thereby indirectly reducing food waste. There is growing evidence that our diet profoundly affects gene expression, which plays an important role in our development, disease response, environmental exposure responses, and aging. A personalized diet customized by individual multi-omics data, daily meals, and routine will improve the quality of life and extend life expectancy. As a new data-driven approach to enhancing food flavor, AI-based computational gastronomy paves the way for a radical change in the food landscape. Understanding the correlation among dishes, ingredients, their molecular components, and flavor perception helps create a flavor printer that generates a molecular mixture with ideal sensory characteristics. By coordinating the relevant experience of food precision nutrition and food intake, AI can serve as a digital diet coach to recommend personalized diets that fit personal health parameters. Computational gastronomy has unlimited possibilities and a bright future for food.

In general, the applications of AI in food flavor prediction, diet regulation, and flavor product design are still very lacking. The use of AI in food flavor should receive more and more attention, and relevant methods should be continuously optimized, and experience should be accumulated to meet the ever-increasing expectations of consumers for food flavor products. Both nutrition and flavor attributes should be taken into account while meeting consumer needs and preferences. AI holds significant potential for transforming food flavor development, yet further research is still needed to refine prediction models and explore broader applications in consumer markets.

AUTHOR CONTRIBUTIONS

Zhiyong Cui: Conceptualization; writing—original draft; writing—review and editing; visualization; validation; methodology; software; data curation; supervision; resources; formal analysis; project administration; investigation. **Chengliang Qi:** Conceptualization. **Tianxing Zhou:** Data curation. **Yanyang Yu:** Validation. **Yueming Wang:** Writing—original draft. **Zhiwei Zhang:**

Writing—original draft. **Yin Zhang**: Writing—original draft. **Wenli Wang**: Writing—review and editing. **Yuan Liu**: Writing—review and editing; funding acquisition; validation.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

ORCID

Yuan Liu  <https://orcid.org/0000-0003-1420-0276>

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